

Electric Potential

Electric Potential

Definition:

Electric potential at a point is the work done in bringing unit positive charge from infinity to that point.

Formula:

$$V = W / q$$

Where:

- V = Electric potential
- W = Work done
- q = Charge

SI Unit of Potential

Volt (V)

Potential Difference

Definition:

Potential difference between two points is the work done in moving unit positive charge from one point to another.

Potential Due to Point Charge

Formula:

$$V = kQ / r$$

Where:

- V = Electric potential
- Q = Source charge
- r = Distance from charge
- k = Electrostatic constant

Equipotential Surface

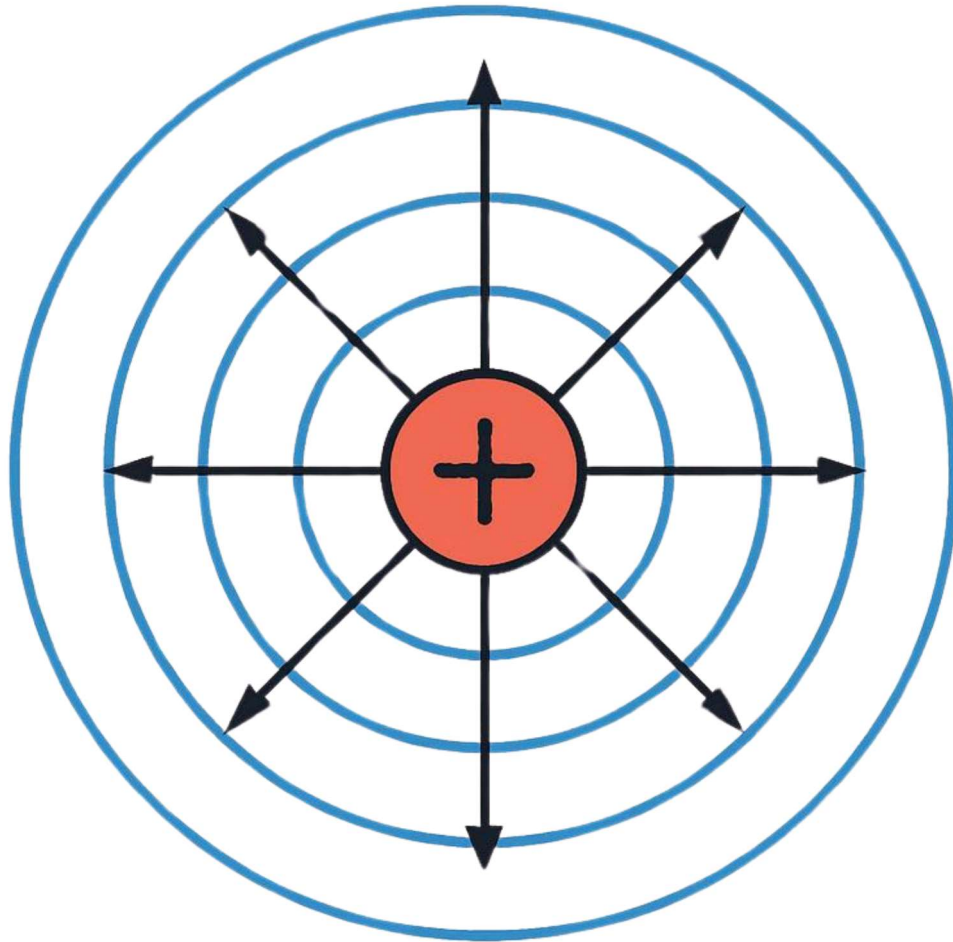
Definition:

A surface on which electric potential remains constant at every point is called equipotential surface.

Properties:

- No work is done while moving charge on an equipotential surface.
- Equipotential surfaces never intersect.
- The electric field is perpendicular to the equipotential surface.

EQUIPOTENTIAL SURFACES



ELECTRIC FIELD LINES

Relation Between Electric Field and Potential

$$E = V / d$$

Where:

- E = Electric field intensity
- V = Potential difference
- d = Distance

Electric Potential Energy

Definition:

The energy possessed by a charge due to its position in an electric field is called electric potential energy.

Formula:

$$U = qV$$

Where:

- U = Potential energy
- q = Charge
- V = Electric potential

Capacitor

Definition:

A device used to store electric charge and electrical energy is called a capacitor.

Capacitance:

$$C = Q / V$$

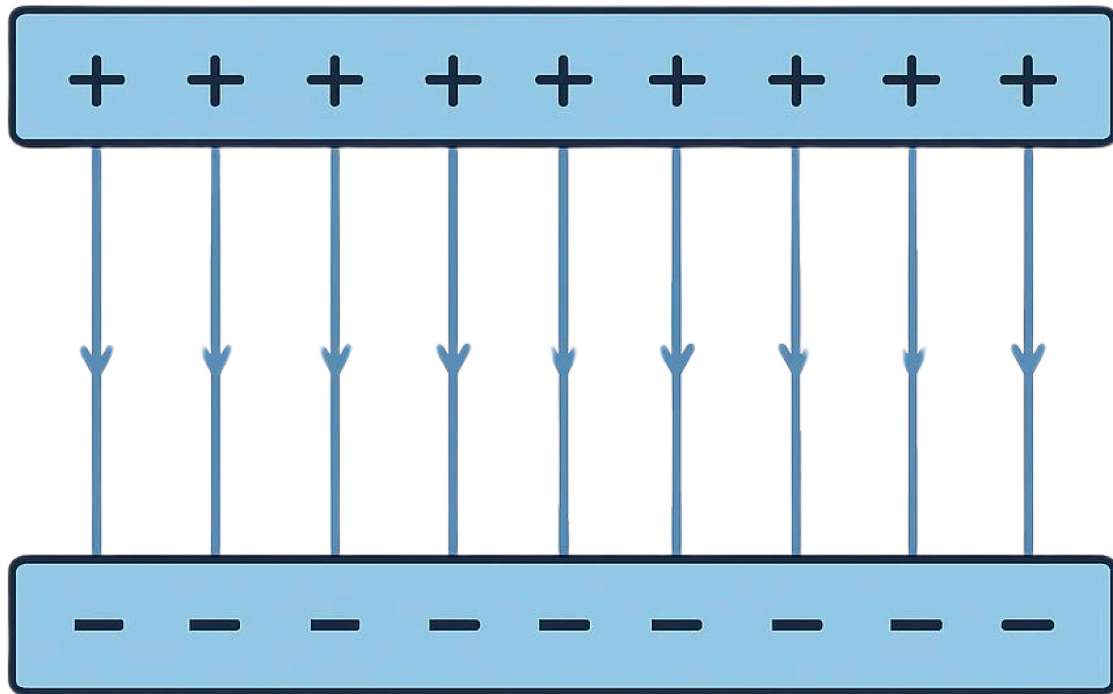
SI Unit:

Farad (F)

Applications of Capacitor

- Electronic circuits
- Camera flash
- Energy storage

PARALLEL PLATE CAPACITOR



Quick Revision Points

- Electric potential is work done per unit charge.
- SI unit of potential is volt.
- Potential due to point charge is inversely proportional to distance.
- Equipotential surfaces have constant potential.
- The electric field is perpendicular to the equipotential surface.
- Potential energy depends on charge and potential.
- Capacitors store electric charge and energy.

Important Formulae

- $V = W / q$
- $V = kQ / r$
- $E = V / d$
- $U = qV$
- $C = Q / V$

Conclusion

Electric potential and capacitance are important concepts in electrostatics and electronic devices.